Harmonic Weighting is Count-Only Inverse-Variance Weighting: A Rigorous Asymptotic Analysis of Intervention Harvesting on Heavy-Tailed Click Logs

Anonymous Author(s)

Abstract

Intervention-harvesting estimators of position bias [1] combine per-cell click-rate estimates across many $(q, d, \text{position})$ cells with widely varying impression counts. We make three contributions. **(i)** We identify *harmonic* weighting $\omega_i = N^i_k N^i_{k'}/(N^i_k + N^i_{k'})$ as the count-only inverse-variance optimum – the classical generalised-least-squares solution of Aitken [2] applied to ULTR. We give an unconditional Cauchy–Schwarz guarantee that harmonic reduces the delta-method variance surrogate $V_k + V_{k'}$ for *any* impression counts, and characterise the exact asymmetric oracle of which harmonic is the $\alpha = \beta$ special case. **(ii)** We replace bootstrap-based variance evaluation – which is provably the wrong instrument on heavy-tailed pair distributions – with the plug-in delta-method asymptotic variance $\widetilde{V}(\omega)$ of the self-normalised weighted ratio estimator [9, 11], and report *asymptotic relative efficiency* (ARE) as the headline finite-sample diagnostic. **(iii)** On the full KDD Cup 2012 Track 2 sponsored-search log (235M impressions; 4.6M interventional pairs), harmonic achieves ARE = 4.53 versus uniform (rising to $\sim 5\times$ on the high-information quartile), with plug-in $\widetilde{V}$ agreeing with empirical click-resample SE to within 1%. A Cochran-Q heterogeneity test on high-information cells rejects PBM at $z \approx 253\sigma$, cleanly separating estimator efficiency from model misspecification. Synthetic (43%–54% MSE reduction at high imbalance) and semi-synthetic Yahoo-LTR (40% variance reduction at $r = 50$) experiments with ground truth corroborate the efficiency claim.

ACM Reference Format:

Anonymous Author(s). 2026. Harmonic Weighting is Count-Only Inverse-Variance Weighting: A Rigorous Asymptotic Analysis of Intervention Harvesting on Heavy-Tailed Click Logs. In *Proceedings of ACM CIKM (CIKM '26)*. ACM, New York, NY, USA, 3 pages. https://doi.org/10.1145/nnnnnnn.nnnnnnn

1 Introduction

Position bias systematically corrupts click signals used for learning-to-rank [6]; intervention harvesting [1] estimates position-bias ratios $R_{k,k'} = p_k/p_{k'}$ from naturally occurring ranking variations without explicit randomisation. On production click logs, the per-cell counts that feed this estimator are extraordinarily heavy-tailed. On the full KDD Cup 2012 Track 2 release (235M impressions; 4.6M interventional pairs at adjacent positions $(1, 2)$), **60.7% of pairs have $N_1 + N_2 \leq 5$ while 0.1% of pairs carry 42% of all impressions** [7]. Public benchmarks hide this shape: Baidu-ULTR [14] has 94% of (q, d, pos) cells with $N = 1$ after deduplication and 78% of adjacent-pair ratios within $[0.5, 2]$.

Equal-weighting heteroscedastic estimators is statistically suboptimal by a century of classical theory: Aitken's generalised-least-squares theorem [2] mandates inverse-variance weights, a result the meta-analysis literature [3, 5] has championed for fifty years. Recent ULTR work [1, 12] introduced *harmonic* weighting $\omega_i = N^i_k N^i_{k'}/(N^i_k + N^i_{k'})$ to reduce variance versus uniform on heavy-tailed counts, but existing validation has relied on *bootstrap* resampling – which we show is provably the wrong instrument for inverse-variance-style weights on singleton-dominated data (Section 3).

Contributions. (1) We identify harmonic as the count-only inverse-variance optimum (Theorem 1), prove an unconditional Cauchy–Schwarz reduction of the delta-method variance surrogate (Theorem 2), and characterise the exact asymmetric oracle of which harmonic is the $\alpha = \beta$ special case (Theorem 3). **(2)** We adopt the plug-in delta-method asymptotic variance $\widetilde{V}(\omega)$ as the appropriate finite-sample efficiency diagnostic on real click logs, replacing bootstrap. **(3)** On the full KDD log we measure $\text{ARE}(\text{harmonic}, \text{uniform}) = 4.53$, calibrated within 1%, and reject PBM at $z \approx 253\sigma$ via Cochran's Q. Synthetic and Yahoo-LTR experiments with ground truth corroborate.

2 Estimator and Theory

Under PBM, $P(\text{click} \mid q, d, k) = p_k\, r_{q,d}$. Given an interventional set $I_{k,k'}$ of (q, d) pairs that appeared at both positions k and k', the position-bias ratio is estimated as [1]

$$\hat{R}_{k,k'} = \frac{A}{B} = \frac{\sum_i \omega_i\, \hat{\theta}^i_k}{\sum_i \omega_i\, \hat{\theta}^i_{k'}}, \quad \hat{\theta}^i_k = \frac{C^i_k}{N^i_k}. \tag{1}$$

This is a self-normalised weighted ratio estimator. Any count-only weight $\omega_i = \omega(N^i_k, N^i_{k'})$ preserves ratio-unbiasedness under PBM [1].

Conditional variance. By the delta method [11], $\text{Var}(\hat{R} \mid \mathbf{N}) \approx R^2[\alpha V_k(\omega) + \beta V_{k'}(\omega)]$ with $V_r(\omega) = \sum_i \omega_i^2/N^i_r/(\sum_i \omega_i)^2$, $\alpha = (1 - p_k)/p_k$, $\beta = (1 - p_{k'})/p_{k'}$.

Harmonic = count-only IVW.. For Bernoulli observations, $\text{Var}(\hat{\theta}^i_k) = \theta^i_k(1 - \theta^i_k)/N^i_k$. With θ treated as constant across cells, the IVW optimum of Aitken [2] for the pair (k, k') is the harmonic mean.

Theorem 1 (Count-only IVW characterisation). *Among all non-negative weights $\omega_i = \omega(N^i_k, N^i_{k'})$, harmonic weighting minimises $V_k(\omega) + V_{k'}(\omega)$.*

Theorem 2 (Unconditional surrogate guarantee). *For any impression counts $\{N^i_k, N^i_{k'}\}_{i=1}^n$, $V_k(\omega^h) + V_{k'}(\omega^h) \leq V_k(\mathbf{1}) + V_{k'}(\mathbf{1})$.*

CIKM '26, Toronto, Canada

ACM ISBN 978-x-xxxx-xxxx-x/YYYY/MM

https://doi.org/10.1145/nnnnnnn.nnnnnnn

SKETCH. Substituting $\omega_i^h = N_k^i N_{k'}^i/(N_k^i + N_{k'}^i)$ collapses the LHS to $1/W$ with $W = \sum_i \omega_i^h$. The RHS is $(1/n^2)\sum_i 1/\omega_i^h$. Cauchy–Schwarz gives $n^2 \le W \sum_i 1/\omega_i^h$. □

THEOREM 3 (EXACT ASYMMETRIC ORACLE). *$\alpha V_k(\omega) + \beta V_{k'}(\omega) \ge (\sum_i 1/s_i)^{-1}$ with $s_i = \alpha/N_k^i + \beta/N_{k'}^i$, attained at $\omega_i^\star \propto N_k^i N_{k'}^i/(\alpha N_{k'}^i + \beta N_k^i)$. Harmonic is the $\alpha = \beta$ special case.*

Theorem 2 is unconditional in the counts – a property that distinguishes harmonic from the min-count heuristic of [1], which has no such guarantee. Theorem 3 provides the oracle benchmark; in synthetic experiments substituting the true α, β into $\omega^\star$ yields variance within ±2% of harmonic.

3 Plug-in Asymptotic Variance: the Right Instrument

$\widetilde{V}$ in closed form. With $A = \sum_i \omega_i \theta_k^i$, $B = \sum_i \omega_i \theta_{k'}^i$, and clicks at k and k' conditionally independent given counts, the delta method gives

$$\widetilde{V}(\omega) = \frac{\sum_i \omega_i^2\, \theta_k^i(1-\theta_k^i)/N_k^i}{A^2} + \frac{\sum_i \omega_i^2\, \theta_{k'}^i(1-\theta_{k'}^i)/N_{k'}^i}{B^2}, \quad (2)$$

$\mathrm{SE}(\hat{R}) = |\hat{R}|\sqrt{\widetilde{V}(\omega)}$. Plugging in $\hat{\theta}_i$ (with light shrinkage toward the marginal) yields the textbook self-normalised IS variance estimator [8–10].

ARE.. For schemes a, b, $\mathrm{ARE}(a,b) = \widetilde{V}(\omega_b)/\widetilde{V}(\omega_a)$: scheme b needs $\mathrm{ARE}(a,b)$ times the data of a for equal precision. $\widetilde{V}$ scales as $1/\mathrm{ESS}(\omega)$, with Kong's effective sample size [8] $\mathrm{ESS}(\omega) = (\sum_i \omega_i)^2/\sum_i \omega_i^2$.

Why bootstrap fails on heavy-tailed click logs. (i) Uniform averages over ∼ 60% near-singleton pairs, producing a smooth bootstrap distribution whose tightness reflects averaging-of-noise, not estimator precision. *(ii)* Pair resampling perturbs the high-information cells that drive any inverse-variance estimator, penalising concentrated weight schemes by construction. *(iii)* Bootstrap conflates estimator efficiency with PBM misspecification; $\widetilde{V}$ does not (Section 4).

Calibration. On KDD at $f = 0.001$, plug-in SE matches empirical click-resample SE within 1% for both uniform (0.1306 vs. 0.1321) and harmonic (0.1113 vs. 0.1120). The plug-in is correct on this data.

4 Real-Log Results: KDD Cup 2012 Track 2

Data. Full release: 149.6M raw rows expanding to 235.6M impressions, 63.8M unique $(q, \mathrm{ad}, \mathrm{pos})$ cells, 8.2M clicks (CTR 3.49%). Pair $(1,2)$ interventional set: 4.61M (q, ad) pairs, 132.3M impressions, median pair size 4, max 2.47M; 60.7% of pairs have $N_1 + N_2 \le 5$.

Headline (Table 1). $\mathrm{ARE}(\mathrm{harmonic}, \mathrm{uniform}) = 4.53$. **Uniform requires ∼ 4.5× the impressions of harmonic for matching precision**, with $\mathrm{SE}_U/\mathrm{SE}_H \approx 2.13$. The min heuristic [1] achieves 4.41, the count-only oracle 4.57; harmonic is the principled parameter-free optimum.

Stratified ARE (Table 2). Schemes tie on noise singletons (where there is nothing to estimate) and harmonic dominates uniform by 5× on the high-information quartile. The gain comes from *not* treating singletons as equal to million-impression cells.

Table 1: Plug-in delta-method asymptotic variance $\widetilde{V}$, SE of $\hat{R}$, ARE vs. uniform, and ESS on full KDD Cup 2012 Track 2 ($n_{\mathrm{pairs}} = 4.61$M). Harmonic, min, and the count-only oracle all reach ARE ≈ 4.5. The full-θ oracle is pathological because per-cell $\hat{\theta}_i$ is too noisy at this scale.

Scheme	$\hat{R}$	SE	ARE	ESS
Uniform	1.683	4.24e-3	1.00	4.61M
Min [1]	1.712	2.05e-3	4.41	2,526
Harmonic	1.710	2.02e-3	**4.53**	1,849
Count-oracle (0.5,1)	1.715	2.02e-3	4.57	1,760
θ-aware oracle	2.343	0.148	0.002	1.1

Table 2: Stratified ARE(harmonic, uniform) by $N_1 + N_2$ quartile on full KDD. Schemes are tied on the noise-dominated quartiles; harmonic dominates ∼ 5× on the high-information quartile where the actual estimation signal lives.

$N_1 + N_2$ quartile	n pairs	$\widetilde{V}_U$	$\mathrm{ARE}(h,u)$
Lowest	1.99M	2.3e-5	1.02
Middle-low	1.38M	1.9e-5	1.09
Highest	1.25M	7.6e-6	**4.98**

Cochran's Q rejects PBM at $z \approx 253\sigma$. On the 16,132 high-information pairs (≥ 5 clicks at each of positions 1 and 2), $Q = 6.17{\times}10^4$, df = 16,131, $Q/\mathrm{df} = 3.83$, $z \approx 253\sigma$. This holds across scales ($z \approx 50$ at $f = 0.1$, $z \approx 7.6$ at $f = 0.01$). PBM rejection explains why schemes converge to slightly different $\hat{R}$ on KDD (uniform 1.68, harmonic 1.71, θ-oracle 2.34): Theorem 1 ratio-unbiasedness [1] requires PBM, which is violated – consistent with prior findings on real logs [12, 13]. The ARE comparison is unaffected as it is a model-internal efficiency statement.

5 Synthetic and Yahoo-LTR Validation

To validate the ARE prediction in settings with ground truth, we run two controlled experiments.

Anchor-item imbalance sweep (synthetic). Following [1], M=10 positions, $p_k = k^{-1}$, anchor-item model with traffic split s and imbalance ratio r, 500 trials. At $s = 0.80$, $r = 100$ harmonic reduces variance by 43.6% vs. uniform, 5% vs. min; reduction grows to ∼ 54% at $s = 0.90$, $r = 50$. Across the (s, r) grid, the data-equivalence factor $\mathrm{DEF} = b_{\mathrm{uni}}/b_{\mathrm{harm}}$ has median 1.31 and maximum 1.94.

Yahoo-LTR semi-synthetic. Same simulator but with real graded relevance from Yahoo Learning to Rank Challenge Set 1 (6936 test queries; clicks simulated via PBM). Harmonic reduces variance by ∼ 40% vs. uniform at imbalance $r = 50$ (Var 0.00015 ⟶ 0.00009; ∼458× lower MSE than aggressive clipping). Production-style traffic simulation with K=5 concurrent rankers (one default at 80% + four treatment at 5% each, swap probability 0.5): under uniform traffic only 0.4% of pairs exceed ratio 10, but under production traffic 16.7% do; harmonic's MSE reduction grows from 6.3% to **17.2%**, confirming the gain scales with traffic-induced imbalance.

6 Conclusion

Framing harmonic weighting as the count-only inverse-variance optimum (Theorems 1–3) and replacing bootstrap with the plug-in delta-method variance $\widetilde{V}$ (Eq. 2) yields rigorous, calibration-verified efficiency comparisons on heavy-tailed real click logs. On the full KDD Cup 2012 Track 2 log, $\mathrm{ARE}(\text{harmonic}, \text{uniform}) = 4.53$, rising to $\sim 5\times$ on information-bearing cells, with PBM heterogeneity quantified separately by Cochran-Q at $z \approx 253\sigma$. Future work: shrinkage variants of the θ-aware oracle (Theorem 3 with $\hat{\theta}_i$ regularised); click models beyond PBM [4, 12, 13] where the intervention-harvesting family must be reformulated.

GenAI Usage Disclosure

The authors used a large-language-model assistant for code generation (KDD aggregation, plug-in variance script, LaTeX table formatting), and for copy-editing. All experimental design, mathematical derivations (Theorems 1–3 and Eq. (2)), interpretation, citations, and final wording were authored and verified by the authors. No LLM-generated content was used without human review. Full code, scripts, and data-processing pipelines are provided as supplementary material.

References

[1] A. Agarwal, X. Wang, C. Li, M. Bendersky, and M. Najork. Estimating position bias without intrusive interventions. In *WSDM*, 2019.
[2] A. C. Aitken. On least squares and linear combination of observations. *Proc. Royal Soc. Edinburgh*, 55:42–48, 1934.
[3] M. Borenstein, L. V. Hedges, J. P. T. Higgins, and H. R. Rothstein. *Introduction to Meta-Analysis.* Wiley, 2009.
[4] A. Chuklin, I. Markov, and M. de Rijke. *Click Models for Web Search.* Morgan & Claypool, 2015.
[5] L. V. Hedges and I. Olkin. *Statistical Methods for Meta-Analysis.* Academic Press, 1985.
[6] T. Joachims, A. Swaminathan, and T. Schnabel. Unbiased learning-to-rank with biased feedback. In *WSDM*, 2017.
[7] Tencent. KDD Cup 2012 Track 2: predict the click-through rate of ads. Competition release, 2012.
[8] A. Kong. A note on importance sampling using standardized weights. Tech. Rep. 348, U. Chicago Statistics, 1992.
[9] A. B. Owen. *Monte Carlo Theory, Methods and Examples*, ch. 9. 2013.
[10] A. Swaminathan and T. Joachims. The self-normalized estimator for counterfactual learning. In *NeurIPS*, 2015.
[11] A. W. van der Vaart. *Asymptotic Statistics*, ch. 3. Cambridge, 1998.
[12] X. Wang, M. Bendersky, D. Metzler, and M. Najork. Position bias estimation for unbiased learning to rank in personal search. In *WSDM*, 2018.
[13] Y. Yue, R. Patel, and H. Roehrig. Beyond position bias: examining result attractiveness as a source of presentation bias. In *WWW*, 2010.
[14] L. Zou et al. A large-scale search dataset for unbiased learning to rank. In *NeurIPS*, 2022.